Linear Functions & Inequalities

Section 1: Slope, Intercepts & Rate of Change

1 A line in the xy-plane passes through the points $(-3, 4)$ and $(5, -8)$. What is the slope of the line?

a. $-\frac{3}{2}$

b. $-\frac{2}{3}$

c. $\frac{3}{2}$

d. -1

2 The linear function f satisfies $f(0) = 8$ and $f(4) = 20$. What is the value of $f(7)$?

a. 26

b. 29

c. 32

d. 35

3 What is the x-intercept of the line given by $4x - 7y = 28$?

a. $(7, 0)$

b. $(4, 0)$

c. $(-7, 0)$

d. $(0, -4)$

Section 2: Parallel and Perpendicular Lines

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4 Line L_1 has the equation $3x + 2y = 10$. Line L_2 is perpendicular to line L_1 and passes through the point $(6, -1)$. What is the equation of line L_2?

a. $y = \frac{2}{3}x - 5$

b. $y = -\frac{2}{3}x + 3$

c. $y = \frac{3}{2}x - 10$

d. $y = -\frac{3}{2}x + 8$

5 Line P has equation $y = 4x - 9$. Line Q is parallel to line P and passes through $(2, 3)$. What is the y-intercept of line Q?

a. -5

b. -3

c. 5

d. 11

6 In the xy-plane, line k passes through $(1, 2)$ and $(4, 8)$. If line m is perpendicular to line k, what is the slope of line m?

a. $-\frac{1}{2}$

b. $\frac{1}{2}$

c. -2

d. 2

Section 3: Single-Variable and Multi-Step Inequalities

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7 What is the greatest integer x that satisfies the inequality $5 - 3x \geq -16$?

a. 6

b. 7

c. 8

d. 9

8 Solve the compound inequality: $-7 \leq 2x - 3 < 11$.

a. $-2 \leq x < 7$

b. $-5 \leq x < 8$

c. $-2 \leq x \leq 7$

d. $-4 \leq x < 7$

9 A cell phone company charges a monthly base fee of 30 dollars plus 0.08 dollars per megabyte of data used over the plan limit. If a subscriber wants their total bill to be at most 54 dollars, what is the maximum number of additional megabytes they can use?

a. 250

b. 300

c. 350

d. 400

Section 4: Two-Variable Inequalities & Systems

10 Which of the following points lies in the solution set of the system $y > 2x - 1$ and $y \leq -x + 6$?

a. $(1, 3)$

b. $(3, 6)$

c. $(0, -2)$

d. $(4, 1)$

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11 The shaded region of an inequality in the xy-plane lies below the solid boundary line $3x - 4y = 12$. Which inequality represents this region?

a. $y \geq \frac{3}{4}x - 3$

b. $y \leq \frac{3}{4}x - 3$

c. $y \leq -\frac{3}{4}x + 3$

d. $y \geq -\frac{3}{4}x + 3$

12 A farmer has at most 120 meters of fencing to enclose a rectangular vegetable garden. If the length must be at least 40 meters, what is the maximum possible width of the garden?

a. 15 meters

b. 20 meters

c. 25 meters

d. 30 meters

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